PID-RS TECHNICAL REPORT SERIES

Concentration for Finite-Alphabet Shared-Exclusions PID under a Dependency Coloring

Finite-sample law control, drift separation, local atom moduli, and a formal evidence boundary

pid-rs project analysis

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Abstract

This document gives project-defined validation, new in `pid-rs`, for the categorical shared-exclusions partial information decomposition (SxPID) defined by Makkeh, Gutknecht, and Wibral. The paper-defined functional is not new. There is no measure called "colored PID": coloring qualifies the sampling theorem and comes from older dependence/concentration machinery, while its composition with SxPID here is project-defined. The document derives a finite-sample concentration bound for the complete empirical joint law when all complete rows share one common finite law and a fixed coloring partitions observations into classes whose complete rows are mutually independent. Dependence across classes can be arbitrary. A class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme improves the ordinary color-count factor for unbalanced classes. A telescoping error allocation gives a time-uniform envelope. A nonidentical-law extension separates random error from explicit drift. A stronger common-support modulus then transfers joint-law error to pointwise and averaged SxPID atoms. The local result includes one-Λ cumulative control, general-source Möbius-row bounds, and complete two-source atom bounds. Exact diamond analysis reduces the synergy modulus to $\Lambda - \eta$. Lean checks deterministic algebraic subclaims, and an executable challenge suite retains counterexamples to weaker assumptions. The result does not validate continuous shared exclusions, arbitrary time series, adaptive evaluation transforms, or binary64 asymptotics.

Contents

1 Status, scope, and provenance

The mathematical functional, the concentration ingredients, and this repository artifact have different origins. Throughout this document, "under a dependency coloring" modifies the observation-dependence contract and concentration proof, not the PID functional. It is not terminology attributed to Makkeh, Gutknecht, or Wibral.

Object	Origin	Status in this document
Categorical SxPID	Paper-defined by Makkeh, Gutknecht, and Wibral; its lattice has a formal-logic basis due to Gutknecht, Wibral, and Makkeh [1, 2]	Implemented for two to four sources; receives a finite-sample bound and an almost-sure exact-real consistency implication when the displayed envelope vanishes
Hoeffding bounded-variable inequality	Paper-defined classical result [3]	Used inside each independent color class
Dependency-color concentration	Published background includes Janson's coloring method and later Hölder formulations [4, 5]	Project-defined proof composition with a class-size proxy that is optimal within the declared Hölder–Hoeffding proof scheme; no novelty claim
Empirical-law L^1 deviation	Published independent and identically distributed (i.i.d.) background includes Weissman et al. [6]	One-sided subset union bound under the stated color contract

Object	Origin	Status in this document
SxPID local modulus	Project-defined validation of the paper-defined functional	One-Λ cumulative control, general-source Möbius-row bounds, and sharper complete two-source atom-specific bounds. The exact ordinary-diamond diameter and an algebraically sharp closed-coordinate conditioned-diamond bound reduce the synergy modulus to $\Lambda - \eta$. Valid full-support SxPID paths have bounded near-attainment evidence, not an exact path-attainment theorem. Endpoint constructions and one bounded law fixture give only local evidence for other coefficients; no result proves global sharpness of the complete atom or averaged bounds
Drift envelope	Project-defined extension	Concentration about the average row law plus explicit bias to a reference law

The machine-readable authority is `method-catalog.json`. The rendered authority is `METHODS.md`. The Rust implementation routes covered by the final composition are the stable categorical functions

`discrete_sxpid2`, `discrete_sxpid2_averaged`, `discrete_sxpid3`, `discrete_sxpid3_averaged`, `discrete_sxpid_n`, and `discrete_sxpid_n_averaged`.

They are exported from `stable::categorical`. This document adds no estimator and no public Rust API. Its proof and executable artifacts are validation evidence.

Reader's guide, foundational notation, and prerequisites

Throughout, log denotes the natural logarithm, and every information quantity is measured in nats.

No prior PID terminology is assumed. A probability law Q on a finite alphabet $\mathcal{Z}$ is a vector of nonnegative cell masses summing to one. Its support is $\operatorname{supp}(Q) = \{z \in \mathcal{Z} : Q(z) > 0\}$. For two laws,

$$\|Q - P\|_1 = \sum_{z \in \mathcal{Z}} |Q(z) - P(z)|, \qquad \mathrm{TV}(Q, P) = \frac{1}{2}\|Q - P\|_1 = \sup_{E \subseteq \mathcal{Z}} |Q(E) - P(E)|.$$

The empirical law assigns each cell its observed relative frequency. Consequently a law-level L^1 bound controls every event probability simultaneously.

For the PID objects, put $[m] = \{1, \dots, m\}$. A nonempty $a \subseteq [m]$ is a source collection, $S_a = (S_i)_{i \in a}$, and

$$\{S_a = s_a\} = \bigcap_{i \in a} \{S_i = s_i\}.$$

A node α is a nonempty antichain of nonempty source collections: no distinct $a, b \in \alpha$ satisfy $a \subset b$. The redundancy order is

$$\alpha \preceq \beta \quad \Longleftrightarrow \quad \text{every } b \in \beta \text{ contains at least one } a \in \alpha.$$

For a realization $z = (s_1, \ldots, s_m, t)$, define

$$A_\alpha = \bigcup_{a \in \alpha} \{S_a = s_a\}, \qquad B_\alpha = A_\alpha \cap \{T = t\}, \qquad C = \{T = t\}.$$

Whenever the displayed event masses are positive, the informative, misinformative, and signed-net cumulative coordinates are

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha), \quad c_\alpha^-(z; Q) = \log \frac{Q(C)}{Q(B_\alpha)}, \quad c_\alpha^{\mathrm{sx}}(z; Q) = c_\alpha^+(z; Q) - c_\alpha^-(z; Q). \tag{1}$$

For $u \in \{+, -, \mathrm{sx}\}$, the pointwise atom π_α^u is the unique finite-poset Möbius inverse:

$$c_\alpha^u(z; Q) = \sum_{\beta \preceq \alpha} \pi_\beta^u(z; Q), \qquad \pi_\alpha^u(z; Q) = \sum_\beta M_{\alpha\beta} c_\beta^u(z; Q). \tag{2}$$

The corresponding law average is

$$\bar{\pi}_\alpha^u(Q) = \sum_{z \in \mathrm{supp}(Q)} Q(z) \pi_\alpha^u(z; Q). \tag{3}$$

This is the SxPID functional to which the later law bound is transferred. Coloring does not alter any object in these two equations; it only supplies a dependence contract for controlling the empirical law.

The two-source dictionary and a worked empirical example. For $m = 2$, the four antichains are

$$\alpha_{\mathrm{red}} = \{\{1\}, \{2\}\}, \qquad \alpha_1 = \{\{1\}\}, \qquad \alpha_2 = \{\{2\}\}, \qquad \alpha_{\mathrm{syn}} = \{\{1, 2\}\}.$$

The covering relations are

$$\alpha_{\mathrm{red}} \prec \alpha_1 \prec \alpha_{\mathrm{syn}}, \qquad \alpha_{\mathrm{red}} \prec \alpha_2 \prec \alpha_{\mathrm{syn}},$$

and α_1, α_2 are incomparable. These nodes name redundancy, unique information from S_1, unique information from S_2, and synergy. For each $u \in \{+, -, \mathrm{sx}\}$,

$$\pi_{\mathrm{red}}^u = c_{\alpha_{\mathrm{red}}}^u, \qquad \pi_1^u = c_{\alpha_1}^u - c_{\alpha_{\mathrm{red}}}^u, \tag{4}$$

$$\pi_2^u = c_{\alpha_2}^u - c_{\alpha_{\mathrm{red}}}^u, \qquad \pi_{\mathrm{syn}}^u = c_{\alpha_{\mathrm{syn}}}^u - c_{\alpha_1}^u - c_{\alpha_2}^u + c_{\alpha_{\mathrm{red}}}^u. \tag{5}$$

For a complete empirical calculation, let Q put mass $1/4$ on each XOR row

$$(S_1, S_2, T) \in \{(0,0,0), (0,1,1), (1,0,1), (1,1,0)\}.$$

Every supported row has $Q(C) = 1/2$ and the following cumulative values:

Node	$Q(A_\alpha)$	$Q(B_\alpha)$	c_α^+	c_α^-	c_α^{sx}
α_{red}	$3/4$	$1/4$	$\log(4/3)$	$\log 2$	$\log(2/3)$
α_1	$1/2$	$1/4$	$\log 2$	$\log 2$	0
α_2	$1/2$	$1/4$	$\log 2$	$\log 2$	0
α_{syn}	$1/4$	$1/4$	$\log 4$	$\log 2$	$\log 2$

Möbius inversion gives

Atom	π^+	π^-	π^{sx}
Redundancy	$\log(4/3)$	$\log 2$	$\log(2/3)$
Unique S_1	$\log(3/2)$	0	$\log(3/2)$
Unique S_2	$\log(3/2)$	0	$\log(3/2)$
Synergy	$\log(4/3)$	0	$\log(4/3)$

All four rows have these same pointwise values, so they are also the averaged atoms. They reconstruct

$$\log(2/3) + 2\log(3/2) + \log(4/3) = \log 2 = I_Q((S_1, S_2); T).$$

The negative redundancy is not clipped; together with either unique atom it gives $I_Q(S_i; T) = 0$.

The concentration proof uses generalized Hölder, Hoeffding's lemma, the exponential Markov/Chernoff argument, the union bound, and Borel–Cantelli. The local-modulus proof uses the fundamental theorem of calculus, finite-poset Möbius inversion, Cauchy–Schwarz, and arithmetic–geometric mean inequalities. These standard results are named at each application; their elementary statements are used but not independently reproved. "Exact real" means mathematical real arithmetic, not a certified error bound for Rust binary64 operations.

2 Finite setting and the dependence contract

Let $Z_1, \ldots, Z_n$ take values in a finite alphabet $\mathcal{Z}$ with $|\mathcal{Z}| = K$. For SxPID, one row is the complete joint symbol

$$Z_i = (S_{1i}, \ldots, S_{mi}, T_i). \tag{6}$$

Dependence between source and target coordinates inside a row is unrestricted. It is the signal that PID analyzes.

For the common-law theorem, assume every row has marginal law P. Let a deterministic color map partition the indices into occupied classes $C_1, \ldots, C_r$. Define

$$n_j = |C_j| > 0, \qquad n = \sum_{j=1}^{r} n_j. \tag{7}$$

The complete rows $\{Z_i : i \in C_j\}$ must be mutually independent for every j. Dependence across different classes can be arbitrary. The color map must not depend on the observed row values. Pairwise independence inside a color is not sufficient. Neither a covariance cutoff nor an unspecified mixing label establishes this mutual-independence contract.

Define

$$\widehat{P}_n(z) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{Z_i = z\}, \qquad V_n = \left(\sum_{j=1}^{r} \sqrt{n_j}\right)^2, \qquad d_{\mathrm{eff},n} = \frac{V_n}{n}. \tag{8}$$

Proposition 2.1 (Effective-color range)**.** *The deterministic proxy obeys*

$$n \le V_n \le rn, \qquad 1 \le d_{\mathrm{eff},n} \le r. \tag{9}$$

Proof. For the lower bound, expand the square of a sum of nonnegative square roots and retain its diagonal terms. For the upper bound, apply Cauchy–Schwarz:

$$\left(\sum_j \sqrt{n_j}\right)^2 \le \left(\sum_j 1^2\right)\left(\sum_j n_j\right) = rn. \tag{10}$$

The Lean artifact proves both inequalities over exact real numbers. □

The upper bound is attained when the occupied color classes have equal sizes. The value $d_{\mathrm{eff},n}$ is a proof constant derived from the declared class sizes. It is not an estimated effective sample size.

3 Finite-sample law concentration

Theorem 3.1 (Dependency-colored empirical-law bound)**.** *Assume $K \ge 2$ and the contract in Section 2. Then, for every $\varepsilon > 0$,*

$$\mathbb{P}\left(\left\|\widehat{P}_n - P\right\|_1 \ge \varepsilon\right) \le \min\left\{1, (2^K - 2)\exp\left(-\frac{n^2\varepsilon^2}{2V_n}\right)\right\}. \tag{11}$$

If at most d colors are occupied, then

$$\mathbb{P}\left(\left\|\widehat{P}_n - P\right\|_1 \ge \varepsilon\right) \le \min\left\{1, (2^K - 2)\exp\left(-\frac{n\varepsilon^2}{2d}\right)\right\}. \tag{12}$$

Proof. Fix a nonempty proper subset $A \subset \mathcal{Z}$. Put

$$Y_i = \mathbf{1}\{Z_i \in A\} - P(A), \qquad S_j = \sum_{i \in C_j} Y_i. \tag{13}$$

Let $R = \sum_j \sqrt{n_j}$ and select the Hölder exponents

$$p_j = \frac{R}{\sqrt{n_j}}, \qquad \sum_j \frac{1}{p_j} = 1. \tag{14}$$

Each $p_j \ge 1$ because $R \ge \sqrt{n_j}$, so these are valid Hölder exponents. For $\lambda > 0$, generalized Hölder gives

$$\mathbb{E}\exp\left(\lambda \sum_j S_j\right) \le \prod_j \left(\mathbb{E}e^{p_j \lambda S_j}\right)^{1/p_j}. \tag{15}$$

Rows are mutually independent inside each color. Hoeffding's lemma for a centered random variable with range length one gives

$$\mathbb{E}e^{p_j\lambda S_j} \le \exp\left(\frac{n_j p_j^2 \lambda^2}{8}\right). \tag{16}$$

Raising the jth bound to $1/p_j$, multiplying, and using $n_j p_j = R\sqrt{n_j}$ gives

$$\prod_j \left[\exp\left(\frac{n_j p_j^2\lambda^2}{8}\right)\right]^{1/p_j} = \exp\left(\frac{\lambda^2}{8}\sum_j n_j p_j\right) \tag{17}$$

$$= \exp\left(\frac{\lambda^2}{8} R \sum_j \sqrt{n_j}\right) = \exp\left(\frac{\lambda^2 V_n}{8}\right). \tag{18}$$

Therefore,

$$\mathbb{E}\exp\left(\lambda\sum_j S_j\right) \le \exp\left(\frac{\lambda^2 V_n}{8}\right). \tag{19}$$

For every $\lambda > 0$, Chernoff's method and Equation (19) give

$$\mathbb{P}\left(\sum_j S_j \geq nt\right) \leq \exp\left(-\lambda nt + \frac{\lambda^2 V_n}{8}\right). \tag{20}$$

The exponent is a convex quadratic. Its derivative $-nt + \lambda V_n/4$ vanishes at $\lambda = 4nt/V_n$; substitution gives $-2n^2t^2/V_n$. Hence

$$\mathbb{P}\Big(\widehat{P}_n(A) - P(A) \geq t\Big) \leq \exp\left(-\frac{2n^2t^2}{V_n}\right). \tag{21}$$

For any signed finite measure with total mass zero,

$$\frac{1}{2}\left\|\widehat{P}_n - P\right\|_1 = \max_{A\subseteq\mathcal{Z}}\{\widehat{P}_n(A) - P(A)\}. \tag{22}$$

Set $t = \varepsilon/2$ and take a union bound over the $2^K - 2$ nonempty proper subsets. This proves Equation (11). Complement subsets already encode the opposite sign, so no additional factor of two is required. Proposition 2.1 gives Equation (12). □

For $K = 1$, the L^1 distance is identically zero, and the finite-sample formulas reduce to the true but empty statement $0 \leq 0$ for every $\varepsilon > 0$. The time-uniform radius contains $\log(2^K - 2)$, so Section 4 requires $K \geq 2$. The subset factor is exponential in K, so the bound can be vacuous on a large alphabet. This result makes no sharpness or sample-complexity claim. If $s = |\operatorname{supp}(P)| \geq 2$ is known before the data are inspected, then $2^s - 2$ can replace $2^K - 2$. An observed support size is not a known population support size. When $s = 1$, the law distance is zero and the result is vacuous.

Proposition 3.2 (Hölder proxy optimum). *For $q_j > 0$ with $\sum_j q_j^{-1} = 1$,*

$$V_n \leq \sum_j n_j q_j. \tag{23}$$

Equality holds for $q_j = R/\sqrt{n_j}$.

Proof. The exponent conditions force $q_j \geq 1$ for every j. Thus, they are valid Hölder exponents. Cauchy–Schwarz gives

$$\left(\sum_j \sqrt{n_j}\right)^2 = \left(\sum_j \sqrt{n_j q_j}\sqrt{q_j^{-1}}\right)^2 \leq \left(\sum_j n_j q_j\right)\left(\sum_j q_j^{-1}\right). \tag{24}$$

The selected exponents make the two Cauchy–Schwarz vectors proportional. □

This is an optimum inside the Hölder–Hoeffding proof scheme for this partition. It is not a claim that V_n is the best possible constant for every admissible joint law.

4 Time-uniform envelope and consistency

Assume $K \geq 2$. For an infinite sequence, require one fixed map from the positive integers to a finite or countable color set. Only finitely many colors can be occupied in each prefix. The entire infinite collection in each color must consist of mutually independent complete rows. Let V_n use the nonempty class sizes in the first n rows. A uniform bound of at most d occupied colors is required only for the coarse radius in Equation (31).

Fix $0 < \alpha < 1$ and define

$$C_K = 2^K - 2, \qquad \alpha_n = \frac{\alpha}{n(n+1)}. \tag{25}$$

The allocation has the exact finite identity

$$\sum_{n=1}^{N} \alpha_n = \alpha\left(1 - \frac{1}{N+1}\right) \le \alpha. \tag{26}$$

Lean proves the unit-scale finite identity by induction. Multiplication by the fixed scalar α gives Equation (26).

Define

$$R_n = \sqrt{\frac{2V_n}{n^2}\log\left(\frac{C_K n(n+1)}{\alpha}\right)}, \tag{27}$$

$$D_n = \min\{2, R_n\}. \tag{28}$$

Theorem 4.1 (Anytime law envelope)**.** *Under the infinite-sequence contract,*

$$\mathbb{P}\left(\forall n \ge 1 : \left\|\widehat{P}_n - P\right\|_1 \le D_n\right) \ge 1 - \alpha. \tag{29}$$

Proof. For every n, substitution in Theorem 3.1 gives

$$C_K \exp\left(-\frac{n^2 R_n^2}{2V_n}\right) = \alpha_n. \tag{30}$$

The Lean artifact checks the algebraic exponent cancellation. Positivity, the square-root substitution, and its probability interpretation remain in this prose proof. A union bound and Equation (26) give simultaneous control by R_n. Every probability-law L^1 distance is also at most two. Thus it is at most $D_n = \min\{2, R_n\}$. ☐

The coarser d-color radius is

$$D_n^{(d)} = \min\left\{2, \sqrt{\frac{2d}{n}\log\left(\frac{C_K n(n+1)}{\alpha}\right)}\right\}. \tag{31}$$

Under one uniform fixed bound d, this is $O(\sqrt{d\log(n)/n})$. For each positive rational error threshold, the tails in Equation (12) are summable. Borel–Cantelli and a countable intersection over those thresholds therefore give

$$\left\|\widehat{P}_n - P\right\|_1 \longrightarrow 0 \quad \text{almost surely}. \tag{32}$$

If the number of colors grows, the displayed envelope converges to zero under the sufficient condition

$$\frac{V_n \log n}{n^2} \longrightarrow 0. \tag{33}$$

This condition is not necessary for empirical-law convergence. For example, i.i.d. rows still converge when a needlessly fine singleton coloring gives $V_n = n^2$. The condition is sufficient for this declared-color envelope. A singleton coloring also permits complete dependence across rows, so the theorem cannot infer convergence from that coloring alone.

Corollary 4.2 (Fixed-width overlapping windows)**.** *Let E_t be i.i.d. innovations. Fix $w \in \mathbb{N}$ before evaluation, and let*

$$Z_t = f(E_t, \ldots, E_{t+w-1}) \tag{34}$$

for one fixed measurable finite-output map f. Coloring row t by $t \bmod w$ satisfies the dependence contract with at most $d = w$ colors.

Proof. Two rows in the same color use disjoint innovation blocks. Every finite collection in that color is therefore mutually independent. The fixed map and the i.i.d. innovation law give the common row law. □

This corollary covers a fixed categorical overlapping-window construction from i.i.d. innovations. It does not cover circular wraparound windows, a width selected from evaluation outcomes, a rolling-start sequence of repeated claims, or innovations with unspecified time dependence.

Definition 4.3 (Strong ℓ-dependence)**.** Let $\ell \in \mathbb{N}_0$. An ordered sequence is strongly ℓ-dependent when

$$\sigma(Z_i : i \leq t) \quad \text{is independent of} \quad \sigma(Z_i : i \geq t + \ell + 1) \tag{35}$$

for every t.

Coloring index t by $t \bmod (\ell+1)$ separates consecutive members of each color by more than ℓ. Repeatedly separate the last row from the sigma-field generated by all preceding rows. This ordered induction proves mutual independence of all complete rows in each color. Pairwise lag independence alone does not prove this property. The common-law theorem still requires one common row law. If row laws differ, the drift theorem and its additional premises apply.

5 Nonidentical row laws and explicit drift

Assume $K \geq 2$. Let row i have law P_i, and keep the within-color mutual-independence contract. Define

$$\overline{P}_n = \frac{1}{n}\sum_{i=1}^{n} P_i. \tag{36}$$

Center the subset indicator for row i by $P_i(A)$. The Hölder–Hoeffding proof is unchanged.

Theorem 5.1 (Average-law and reference-law bounds)**.** *For every $\varepsilon > 0$,*

$$\mathbb{P}\Big(\Big\|\widehat{P}_n - \overline{P}_n\Big\|_1 \geq \varepsilon\Big) \leq \min\left\{1, C_K \exp\left(-\frac{n^2\varepsilon^2}{2V_n}\right)\right\}. \tag{37}$$

For an explicit reference law $P_\star$, put

$$b_n = \Big\|\overline{P}_n - P_\star\Big\|_1. \tag{38}$$

Then

$$\mathbb{P}\Big(\Big\|\widehat{P}_n - P_\star\Big\|_1 \geq b_n + \varepsilon\Big) \leq \min\left\{1, C_K \exp\left(-\frac{n^2\varepsilon^2}{2V_n}\right)\right\}. \tag{39}$$

Proof. Equation (37) follows from the row-specific centering. The triangle inequality gives

$$\Big\|\widehat{P}_n - P_\star\Big\|_1 \leq \Big\|\widehat{P}_n - \overline{P}_n\Big\|_1 + b_n, \tag{40}$$

which proves Equation (39). □

If $\|P_i - P_\star\|_1 \le \beta_i$, then convexity gives

$$b_n \le \frac{1}{n}\sum_{i=1}^{n}\beta_i. \tag{41}$$

Under one fixed infinite row-law sequence and coloring with the same premises, the all-prefix reference-law envelope is

$$\mathbb{P}\Big(\forall n \ge 1 :\ \Big\|\widehat{P}_n - P_\star\Big\|_1 \le \min\{2, b_n + D_n\}\Big) \ge 1-\alpha. \tag{42}$$

The bias term is necessary. Concentration about $\overline{P}_n$ does not identify a fixed scientific estimand.

6 Strengthened common-support local SxPID continuity

This section records claim `SXPID-TV-001`, revision 2. It gives project-defined validation of a paper-defined functional. It does not define a new PID measure or estimator. All statements use exact real arithmetic.

Claim field	Auditable content
Objects	Two laws on one fixed complete finite source–target alphabet and the complete redundancy lattice
Assumptions	Support inclusion, the strict L^1 margin in Equation (44), a positive population cell floor, and exact-real arithmetic; the first two conditions imply equal support
Conclusion	General-source and sharper complete two-source pointwise and averaged moduli stated below
Evidence	Prose proof, machine-checked deterministic subclaims, a high-precision oracle, and a bounded Rust comparison
Not established	A binary64 theorem, estimator calibration, continuous-PID validity, or complete formal refinement

Let p and q be laws on one fixed complete joint alphabet. Define

$$\delta = \|q-p\|_1\,, \qquad \eta = \frac{\delta}{2}, \qquad p_{\min} = \min_{z\in\mathrm{supp}(p)} p(z), \qquad L = \log(1/p_{\min}). \tag{43}$$

Assume

$$\mathrm{supp}(q) \subseteq \mathrm{supp}(p), \qquad 0 \le \delta < 2p_{\min}. \tag{44}$$

The convention $\eta = \sup_E |q(E) - p(E)|$ is total variation. Equation (44) implies equal support: for each $z \in \mathrm{supp}(p)$, the singleton-event bound gives $q(z) \ge p(z) - \eta \ge p_{\min} - \eta > 0$, while the assumed inclusion gives the reverse direction. Put

$$p_s = (1-s)p + sq, \qquad \mu_s = p_{\min} - s\eta, \qquad 0 \le s \le 1. \tag{45}$$

For $z \in \mathrm{supp}(p)$, the same singleton bound along the segment gives $p_s(z) \ge p(z) - s\eta \ge \mu_s > 0$. Thus every supported cell under p_s has the stated floor.

Lemma 6.1 (Path oscillation)**.** *Let $\Delta = q - p$. For every real-valued function g on the finite common support,*

$$\left|\sum_z \Delta(z)g(z)\right| \le \eta\left(\max_z g(z) - \min_z g(z)\right). \tag{46}$$

If a differentiable functional F satisfies $\operatorname{osc}\nabla F(p_s) \le 1/\mu_s$*, then*

$$|F(q) - F(p)| \le \int_0^1 \frac{\eta}{p_{\min} - s\eta}\, ds = \log \frac{p_{\min}}{p_{\min} - \eta} =: \Lambda(\delta, p_{\min}). \tag{47}$$

Proof. Subtract the midpoint of the range of g in Equation (46) and use $\sum_z \Delta(z) = 0$. Apply that inequality to the directional derivative of F along p_s. An empty event region carries no signed mass, so it need not be included among the attained gradient values when taking the maximum and minimum. The strict support margin makes the complete segment positive, so the fundamental theorem of calculus and direct integration give Equation (47). □

Fix a supported realization $z = (s_1, \ldots, s_m, t)$ and a nonempty antichain α of nonempty source subsets. Define

$$A_\alpha = \bigcup_{a\in\alpha} \{S_a = s_a\}, \qquad B_\alpha = A_\alpha \cap \{T = t\}, \qquad C = \{T = t\}. \tag{48}$$

By definition, $B_\alpha = A_\alpha \cap C$, and the keyed realization belongs to B_α. For a law Q, define

$$c_\alpha^+(z; Q) = -\log Q(A_\alpha), \tag{49}$$

$$c_\alpha^-(z; Q) = \log \frac{Q(C)}{Q(B_\alpha)}, \tag{50}$$

$$c_\alpha^{\mathrm{sx}}(z; Q) = \log \frac{Q(B_\alpha)}{Q(A_\alpha)Q(C)}. \tag{51}$$

Proposition 6.2 (One-Λ cumulative modulus)**.** *For* $u \in \{+, -, \mathrm{sx}\}$,

$$|c_\alpha^u(z; q) - c_\alpha^u(z; p)| \le \Lambda. \tag{52}$$

Proof. Write $A = A_\alpha$ and $B = B_\alpha$. The cell-gradient values are

Region	∇c^+	∇c^-	∇c^{sx}
B	$-1/Q(A)$	$1/Q(C) - 1/Q(B)$	$1/Q(B) - 1/Q(A) - 1/Q(C)$
$A \setminus C$	$-1/Q(A)$	0	$-1/Q(A)$
$C \setminus A$	0	$1/Q(C)$	$-1/Q(C)$
outside $A \cup C$	0	0	0

The informative column takes only $-1/Q(A)$ and zero, so its diameter is $1/Q(A) \le 1/Q(B)$. In the misinformative column, the B value is nonpositive, the $C \setminus A$ value is nonnegative, and their difference is exactly $1/Q(B)$; deleting an empty region can only reduce the diameter.

For completeness, write the four net-column values as

$$v_B = \frac{1}{Q(B)} - \frac{1}{Q(A)} - \frac{1}{Q(C)}, \quad v_A = -\frac{1}{Q(A)}, \quad v_C = -\frac{1}{Q(C)}, \quad v_O = 0.$$

The six unordered differences are

$$v_B - v_A = \frac{1}{Q(B)} - \frac{1}{Q(C)}, \qquad v_B - v_C = \frac{1}{Q(B)} - \frac{1}{Q(A)}, \tag{53}$$

$$v_B - v_O = \frac{1}{Q(B)} - \frac{1}{Q(A)} - \frac{1}{Q(C)}, \qquad v_A - v_C = \frac{1}{Q(C)} - \frac{1}{Q(A)}, \tag{54}$$

$$v_A - v_O = -\frac{1}{Q(A)}, \qquad v_C - v_O = -\frac{1}{Q(C)}. \tag{55}$$

Because $Q(B) \le Q(A), Q(C)$, the absolute value of each difference is at most $1/Q(B)$: for the third one, both $1/Q(A)$ and $1/Q(C)$ lie in $[0, 1/Q(B)]$, so its value lies in $[-1/Q(B), 1/Q(B)]$. Since $p_s(B) \ge \mu_s$, Lemma 6.1 proves the result. This direct net calculation avoids the nonoptimal factor of two obtained by adding separate component bounds. □

Let M be the fixed Möbius matrix of the complete redundancy lattice. Define

$$\pi^u_\alpha(z;Q) = \sum_\beta M_{\alpha\beta} c^u_\beta(z;Q), \qquad r_\alpha = \sum_\beta |M_{\alpha\beta}|, \tag{56}$$

$$\overline{\pi}^u_\alpha(Q) = \sum_{z\in\mathrm{supp}(Q)} Q(z)\pi^u_\alpha(z;Q). \tag{57}$$

Also define

$$h = \log\frac{2}{1+p_{\min}}, \qquad J = L - 2h \geq 0. \tag{58}$$

Theorem 6.3 (General-source atom bounds)**.** *For every complete-lattice node and* $u \in \{+,-,\mathrm{sx}\}$,

$$|\pi^u_\alpha(z;q) - \pi^u_\alpha(z;p)| \leq r_\alpha\Lambda. \tag{59}$$

The averaged component bounds are

$$\left|\overline{\pi}^\pm_\alpha(q) - \overline{\pi}^\pm_\alpha(p)\right| \leq \min\left\{L,\ r_\alpha\Lambda + \eta L\right\}. \tag{60}$$

The averaged net bound is

$$|\overline{\pi}^{\mathrm{sx}}_\alpha(q) - \overline{\pi}^{\mathrm{sx}}_\alpha(p)| \leq \min\left\{2L, r_\alpha\Lambda + \delta\min\{L, r_\alpha(L-h)\}\right\}. \tag{61}$$

Proof. Equation (59) follows from Proposition 6.2 and the triangle inequality. Makkeh, Gutknecht, and Wibral prove that every informative and misinformative pointwise atom is nonnegative on the complete finite antichain lattice [1, Theorem IV.3 and Appendix A]. Their base-2 sign result is unchanged after conversion to nats. A positive-probability keyed realization is enough for that published theorem; full support of the complete table is not required.

Let $\top = \{[m]\}$ be the top node. Its source event is the realized full-source tuple. For a supported $z = (s_{[m]}, t)$,

$$c^+_\top(z;p) = -\log p(S_{[m]} = s_{[m]}), \tag{62}$$

$$c^-_\top(z;p) = \log\frac{p(T=t)}{p(S_{[m]} = s_{[m]}, T=t)} = -\log p(S_{[m]} = s_{[m]} \mid T=t). \tag{63}$$

The two probabilities inside the negative logarithms are at least $p(z) \geq p_{\min}$: the marginal contains the keyed cell, and $p(S_{[m]} = s_{[m]} \mid T=t) = p(z)/p(T=t) \geq p(z)$. Since every lattice node lies below $\top$,

$$c^\pm_\top(z;p) = \sum_\beta \pi^\pm_\beta(z;p).$$

Component nonnegativity therefore implies, for every α,

$$0 \leq \pi^+_\alpha, \pi^-_\alpha \leq L, \qquad -L \leq \pi^{\mathrm{sx}}_\alpha \leq L. \tag{64}$$

For a cumulative net value, write $a = p(A_\alpha)$, $b = p(B_\alpha)$, and $c = p(C)$. Then $a, c \geq b \geq p_{\min}$ and $a + c - b \leq 1$. Thus

$$ac \leq \frac{(a+c)^2}{4} \leq \frac{(1+b)^2}{4}, \tag{65}$$

while $ac \geq b^2$. Consequently,

$$\frac{4b}{(1+b)^2} \leq \frac{b}{ac} \leq \frac{1}{b}. \tag{66}$$

The function $4b/(1+b)^2$ is nondecreasing on $[0,1]$, and $4p_{\min}/(1+p_{\min})^2 = e^{-J}$. Taking logarithms gives

$$-J \leq \log\frac{b}{ac} \leq L. \tag{67}$$

The cumulative range width is $2(L-h)$. A Möbius row gives net-atom oscillation at most $2r_\alpha(L-h)$; Equation (64) also gives oscillation at most $2L$.

Split an average change exactly as

$$\left|\sum_z q(z)\pi^u_\alpha(z;q) - \sum_z p(z)\pi^u_\alpha(z;p)\right| \le \sum_z q(z)|\pi^u_\alpha(z;q) - \pi^u_\alpha(z;p)| \tag{68}$$

$$+\left|\sum_z (q(z)-p(z))\pi^u_\alpha(z;p)\right|. \tag{69}$$

The first term is at most $r_\alpha\Lambda$. Center the second term as in Equation (46). A component atom has range width at most L, so its weight term is at most ηL. A net atom has range width at most both $2L$ and $2r_\alpha(L-h)$, so its weight term is at most $\delta\min\{L, r_\alpha(L-h)\}$. This proves the inner bounds.

For the outer caps, let $D = |\operatorname{supp}(p)| = |\operatorname{supp}(q)|$. Since every p-cell has mass at least $p_{\min}$, $D \le 1/p_{\min}$. Write $H_\rho(X) = -\sum_x \rho(x)\log\rho(x)$ and use the usual conditional entropy. For either law $\rho \in \{p,q\}$, averaging the two top-node surprisals gives

$$\sum_\beta \overline{\pi}^+_\beta(\rho) = H_\rho(S_{[m]}), \qquad \sum_\beta \overline{\pi}^-_\beta(\rho) = H_\rho(S_{[m]} \mid T).$$

Component nonnegativity and the finite-support entropy bounds imply

$$H_\rho(S_{[m]}) \le \log D, \qquad H_\rho(S_{[m]} \mid T) \le H_\rho(S_{[m]}, T) \le \log D \le L.$$

Thus every averaged component atom lies in $[0, L]$, every averaged net atom lies in $[-L, L]$, and differences under the two laws are capped by L and $2L$, respectively. $\square$

Let $\mathsf{Z}_{\alpha\beta} = \mathbf{1}\{\beta \preceq \alpha\}$ be the zeta matrix and let $\perp$ be the bottom node. Since $\mathsf{Z}e_\perp = \mathbf{1}$, inversion gives $M\mathbf{1} = e_\perp$. Hence the bottom row of M is its unit row and has absolute row sum one, whereas every nonbottom row sums to zero. A nonbottom row is a nonzero integer vector with diagonal entry one; zero row sum therefore forces at least one positive and one negative entry, so its absolute row sum is at least two. Consequently the bottom row has $r_\alpha = 1$ and net weight coefficient $L-h$. Since $J = L - 2h \ge 0$, every nonbottom row has $r_\alpha \ge 2$, so its inner net coefficient simplifies to L.

Theorem 6.4 (Complete two-source improvement)**.** *Define*

$$\Lambda_{\text{syn}} = \log\frac{p_{\min}}{p_{\min}-\eta} - \eta = \Lambda - \eta. \tag{70}$$

For redundancy and either unique atom, and for $u \in \{+, -, \text{sx}\}$,

$$|\pi^u_\alpha(z;q) - \pi^u_\alpha(z;p)| \le \Lambda. \tag{71}$$

For synergy and the same three component labels,

$$|\pi^u_{\text{syn}}(z;q) - \pi^u_{\text{syn}}(z;p)| \le \Lambda_{\text{syn}}. \tag{72}$$

Put

$$J_q = \log\frac{(1+p_{\min}-\eta)^2}{4(p_{\min}-\eta)}. \tag{73}$$

The averaged bounds are

Atom	*Component average change*	*Net average change*
Redundancy	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L-h)\}$
Either unique atom	$\min\{L, \Lambda + \eta L\}$	$\min\{2L, \Lambda + \delta(L-h)\}$
Synergy	$\min\{L, J_q, \Lambda_{\text{syn}} + \eta J\}$	$\min\{2L, J + J_q, \Lambda_{\text{syn}} + \delta J\}$

Proof. Let $E_i = \{S_i = s_i\}$, $U = E_1 \cup E_2$, and $C = \{T = t\}$. Redundancy uses one event log, one nested log ratio, or one intersection-PMI expression, so Proposition 6.2 applies.

For one unique atom, put

$$N(a, b) = \log \frac{a+b}{a}. \tag{74}$$

Its two nonzero gradient values are

$$D_a = \frac{1}{a+b} - \frac{1}{a}, \qquad D_b = \frac{1}{a+b}, \tag{75}$$

whose diameter is $1/a$. For a path law $Q = p_s$, put

$$x_a = Q(E_i \cap C), \qquad x_b = Q((U \setminus E_i) \cap C), \tag{76}$$

$$y_a = Q(E_i \cap C^{\mathsf{c}}), \qquad y_b = Q((U \setminus E_i) \cap C^{\mathsf{c}}). \tag{77}$$

The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise. Define

$$G_N(x, y) = N(x+y) - N(x). \tag{78}$$

The two-source inversion identifies G_N with the signed unique atom. Indeed, $\pi_i^u = c^u_{\alpha_i} - c^u_{\alpha_{\text{red}}}$ and $U = E_i \,\dot\cup\, (U \setminus E_i)$. Hence

$$\pi_i^+(z; Q) = \log \frac{Q(U)}{Q(E_i)} = N(x_a + y_a, x_b + y_b), \tag{79}$$

$$\pi_i^-(z; Q) = \log \frac{Q(U \cap C)}{Q(E_i \cap C)} = N(x_a, x_b), \tag{80}$$

$$\pi_i^{\text{sx}}(z; Q) = N(x_a + y_a, x_b + y_b) - N(x_a, x_b) = G_N(x, y). \tag{81}$$

Thus $N(x+y)$ is the informative unique component, $N(x)$ the misinformative component, and $G_N(x, y)$ their signed difference.

For $i \in \{a, b\}$, the y_i-gradient coordinate is $F_i = D_i(x+y)$ and the x_i-gradient coordinate is $X_i = F_i - D_i(x)$. A coordinate outside the two event regions has gradient value zero.

Put

$$A = \frac{1}{x_a}, \qquad B = \frac{1}{x_a + x_b}, \qquad E = \frac{1}{x_a + y_a}, \qquad P = \frac{1}{x_a + x_b + y_a + y_b}. \tag{82}$$

The five values are

$$F_a = P - E, \qquad F_b = P, \qquad X_a = P - E - B + A, \qquad X_b = P - B, \qquad 0. \tag{83}$$

Their minimum is $\min\{F_a, X_b\}$, their maximum is $\max\{F_b, X_a\}$, and their exact diameter is

$$\max\{E,\, B,\, A - B,\, A - E\}. \tag{84}$$

The full-union reciprocal P cancels. Each displayed term lies in $[0, A]$, so the diameter is at most $1/x_a$. Equivalently, the F_a–F_b difference is at most $1/(x_a + y_a) \leq 1/x_a$, and the same-index differences $F_i - X_i = D_i(x)$ have magnitude at most $1/x_a$. Also,

$$X_a - X_b = \frac{1}{x_a} - \frac{1}{x_a + y_a}, \tag{85}$$

and the two crossed differences are

$$F_a - X_b = -\frac{1}{x_a + y_a} + D_b(x), \qquad F_b - X_a = \frac{1}{x_a + y_a} + D_a(x). \tag{86}$$

Each crossed difference is a difference of two values in $[0, 1/x_a]$. Each of F_a, F_b, X_a, X_b has magnitude at most $1/x_a$, which also controls every pair with the outside zero coordinate. Thus all five gradient values have diameter at most $1/x_a$.

Lean proves the candidate extrema, Equation (84), pairwise domination, and attainment. It also proves that no positive quantity can be subtracted uniformly from $1/x_a$ over the nonnegative algebraic mass domain. Its boundary witness sets $x_b = y_a = y_b = 0$ and compares the full-event exclusive coordinate with the outside coordinate. This zero-side-mass construction is an algebraic gradient witness. It is not by itself a supported common-law perturbation or an SxPID-realizability result. The realizable two-cell atom counterexamples below separately show that unique information can attain the unrefined Λ modulus.

For synergy, partition the source space into $E_1 \cap E_2$, $E_1 \setminus E_2$, $E_2 \setminus E_1$, and the outside region $O = (E_1 \cup E_2)^{\mathsf{c}}$. For any law, let a, b, c, o be the masses of these four regions in that order. Define

$$\Phi(a, b, c) = \log \frac{(a + b)(a + c)}{a(a + b + c)}. \tag{87}$$

Its gradient values, including the outside region o, are

$$D_a = \frac{1}{a+b} + \frac{1}{a+c} - \frac{1}{a} - \frac{1}{a+b+c}, \tag{88}$$

$$D_b = \frac{1}{a+b} - \frac{1}{a+b+c}, \qquad D_c = \frac{1}{a+c} - \frac{1}{a+b+c}, \qquad D_o = 0. \tag{89}$$

In particular,

$$D_a = -\frac{bc(2a + b + c)}{a(a+b)(a+c)(a+b+c)} \le 0,$$

while $D_b, D_c \ge 0$. Moreover,

$$D_b - D_a = \frac{1}{a} - \frac{1}{a+c}, \qquad D_c - D_a = \frac{1}{a} - \frac{1}{a+b}. \tag{90}$$

Since

$$D_b - D_c = \frac{1}{a+b} - \frac{1}{a+c},$$

if $b \le c$, then D_b is the maximum and D_a is the minimum. If $c \le b$, then D_c is the maximum and D_a is the minimum. Therefore,

$$\operatorname{osc} D = \frac{1}{a} - \frac{1}{a + \max\{b, c\}}. \tag{91}$$

For a probability partition, $a + \max\{b, c\} \le a + b + c \le 1$, so

$$\operatorname{osc} D \le \frac{1}{a} - 1. \tag{92}$$

The informative synergy component is a diamond function on the full source-region masses. The misinformative component is a diamond function on nonnegative target-restricted region masses whose total is at most one.

For a path law $Q = p_s$, define

$$(x_a, x_b, x_c, x_o) = (Q(E_1 \cap E_2 \cap C), Q((E_1 \setminus E_2) \cap C), Q((E_2 \setminus E_1) \cap C), Q(O \cap C)), \tag{93}$$

and define (y_a, y_b, y_c, y_o) by replacing C with C^{c}. The keyed cell gives $x_a \geq \mu_s > 0$. Addition is componentwise, and

$$G_\Phi(x, y) = \Phi(x + y) - \Phi(x). \tag{94}$$

For $i \in \{a, b, c, o\}$, the y_i-gradient coordinate is $F_i = D_i(x + y)$ and the x_i-gradient coordinate is $X_i = F_i - D_i(x)$. These are the eight gradient values.

The ordinary-diamond signs give

$$F_a \leq 0 \leq F_b, F_c, \qquad X_a \geq F_a, \qquad X_b \leq F_b, \qquad X_c \leq F_c. \tag{95}$$

Both outside coordinates are zero. Hence

$$\operatorname{diam}(F, X) = \max\{F_b, F_c, X_a\} - \min\{F_a, X_b, X_c\}. \tag{96}$$

Put

$$A = \frac{1}{x_a}, \qquad B = \frac{1}{x_a + x_b}, \qquad C = \frac{1}{x_a + x_c}, \qquad D = \frac{1}{x_a + x_b + x_c}, \tag{97}$$

$$E = \frac{1}{x_a + y_a}, \quad P = \frac{1}{x_a + x_b + y_a + y_b}, \quad Q = \frac{1}{x_a + x_c + y_a + y_c}, \quad R = \frac{1}{S}, \tag{98}$$

where $S = x_a + x_b + x_c + y_a + y_b + y_c$. The nine candidate maximum-minus-minimum differences are

$$F_b - F_a = E - Q, \qquad F_c - F_a = E - P, \qquad X_a - F_a = A + D - B - C, \tag{99}$$

$$F_b - X_b = B - D, \qquad F_c - X_c = C - D, \qquad X_a - X_b = A - C + Q - E, \tag{100}$$

$$X_a - X_c = A - B + P - E, \quad F_b - X_c = P - Q + C - D, \quad F_c - X_b = Q - P + B - D. \tag{101}$$

The reciprocal orders are

$$R \leq P, Q \leq E \leq A, \qquad P \leq B, \quad Q \leq C, \quad D \leq B, C \leq A, \tag{102}$$

and $R \leq D$. Reciprocal supermodularity on the base diamond is an algebraic consequence, not an extra premise. With $a = x_a > 0$, $b = x_b \geq 0$, and $c = x_c \geq 0$,

$$A + D - B - C = \frac{1}{a} + \frac{1}{a + b + c} - \frac{1}{a + b} - \frac{1}{a + c} \tag{103}$$

$$= \frac{bc(2a + b + c)}{a(a + b)(a + c)(a + b + c)} \geq 0. \tag{104}$$

Thus $B + C \leq A + D$. The first seven candidate differences obey

$$\begin{aligned} &E - Q \leq A - R, && E - P \leq A - R, \quad A + D - B - C \leq A - R, \\ &B - D \leq A - R, && C - D \leq A - R, \quad A - C + Q - E \leq A - C \leq A - R, \\ &A - B + P - E \leq A - B \leq A - R. \end{aligned}$$

For the first inequality on the second line, for example, $B \leq A$ and $D \geq R$; the other direct comparisons use the displayed reciprocal order. For the crossed difference $F_b - X_c$, if $P \leq Q$, then

$$P - Q + C - D \leq C - D \leq A - R.$$

If $Q < P$, then

$$P - Q + C - D \leq B - Q + C - D \leq A - Q \leq A - R,$$

where the middle inequality is $B + C \le A + D$. The other crossed difference follows by exchanging (B, P) with (C, Q). Therefore,

$$\operatorname{diam}(F, X) \le \frac{1}{x_a} - \frac{1}{S}. \tag{105}$$

All nine extremal regimes in Equation (96) can occur, even when all six displayed masses are positive. There is no one fixed-pair formula. The universal bound in Equation (105) is attained on the closed nonnegative algebraic mass domain. This statement concerns the eight gradient coordinates; it is not yet an attainment statement for a support-preserving SxPID path.

Every probability assignment has $S \le 1$. Since the keyed cell gives $x_a \ge \mu_s$, Equations (92) and (105) bound each synergy gradient diameter by $1/\mu_s - 1$. Centered total variation and path integration give

$$\eta \int_0^1 \left(\frac{1}{p_{\min} - s\eta} - 1 \right) ds = \Lambda_{\text{syn}}. \tag{106}$$

This proves Equation (72). The redundancy and unique calculations above prove Equation (71). The synergy modulus is nonnegative because $-\log(1 - \eta/p_{\min}) \ge \eta/p_{\min} \ge \eta$.

The synergy improvement cannot be applied to the other atoms. Start with two cells of mass $1/2$ and move mass η from the first to the second. Redundancy informative and net use cells $(0, 0, 0)$ and $(1, 1, 1)$; redundancy misinformative uses $(0, 0, 0)$ and $(1, 1, 0)$. Source-1 unique informative and net use $(1, 1, 1)$ and $(0, 1, 0)$; its misinformative component uses $(1, 1, 1)$ and $(0, 1, 1)$. Exchange the sources for source-2 unique information. In each case, the stated first-cell atom is $-\log p(z)$, so its change is exactly $\log[(1/2)/(1/2 - \eta)] = \Lambda > \Lambda_{\text{syn}}$.

Formal boundary, counterexample, and sharpness. Lean proves the exact ordinary-diamond diameter and an attaining ordered pair with

`ordinary_diamond_gradient_exact_coordinate_diameter_le`
`ordinary_diamond_gradient_exact_coordinate_diameter_attained`.

It proves that the candidate extrema in Equation (96) are attained and give the exact eight-coordinate conditioned-diamond diameter with

`conditioned_diamond_lifted_gradient_exact_coordinate_diameter_le`
`conditioned_diamond_lifted_gradient_exact_coordinate_diameter_attained`.

The theorem `conditioned_diamond_lifted_gradient_refined_coordinate_diameter_le` checks Equation (105) over the exact real numbers by expanding all 64 ordered coordinate pairs. It assumes only

$$x_a > 0, \qquad x_b, x_c, y_a, y_b, y_c \ge 0. \tag{107}$$

It needs no positive exclusive-region mass, outside-region mass, normalization, or upper bound on the displayed masses. The theorem `conditioned_diamond_lifted_gradient_refined_bound_attained` proves equality for $x = (x_a, 0, 0)$ and $y = (0, 0, S - x_a)$ when $0 < x_a \le S$. Neither theorem formalizes differentiation, path integration, probability events, or the identification with net SxPID synergy. The older declaration `conditioned_diamond_lifted_gradient_coordinate_diameter_le` remains as the coarser $1/x_a$ corollary. The following declarations check the $1/x_a - 1$ pairwise and absolute-coordinate corollaries when the six displayed masses total at most one:

`conditioned_diamond_lifted_gradient_probability_domain_coordinate_diameter_le`
`conditioned_diamond_lifted_gradient_probability_domain_absolute_coordinate_le`

Separate validity of the base and full diamonds is insufficient if a negative exclusive complement-region lift mass is allowed. Let

$$x = (1/4, 1/4, 0), \qquad y = (0, -1/4, 1/2). \tag{108}$$

The base and full vectors x and $x + y = (1/4, 0, 1/2)$ are separately valid diamonds, but $y_b < 0$. The lifted coordinates include $F_b = 8/3$ and $X_c = -2$. Their difference is $14/3$, which exceeds both the endpoint-only refined claim $8/3$ and the coarse claim $1/x_a = 4$. The exact fixture retains this counterexample.

The fixture also retains the source-exchanged negative-exclusive case. A third case uses

$$x = (1/2, 0, 0), \qquad y = (-1/4, 3/4, 0). \tag{109}$$

The base and full diamonds are valid, but the coordinate diameter is $3 > 2 = 1/x_a$ and also exceeds the endpoint-only refined claim 1. Thus separate endpoint validity does not replace the componentwise nonnegative lift premise.

The refined coefficient is attained. For

$$x = (\varepsilon, 0, 0), \qquad y = (0, 0, 1-\varepsilon), \qquad 0 < \varepsilon < 1, \tag{110}$$

the displayed mass total is one, and $F_b - X_c = (1-\varepsilon)/\varepsilon$ equals $1/x_a - 1/S$. Its ratio to the refined bound is one. Its ratio to the older $1/x_a$ bound is $1-\varepsilon$. The fixture uses $\varepsilon = 1/1000$ and the source-swapped case. The Python generator and the Rust test independently reconstruct all eight coordinates, the refined bound, and all 64 ordered differences in each of seven rational cases. Nine additional assignments have six positive displayed masses that sum to one and realize all nine extremal regimes. The checks also reconstruct the exact ordinary-diamond and conditioned-nested identities on the same inputs. One case has zero lift. One algebra-only case has displayed mass total 41 and checks that normalization is not used by the algebraic theorem. It is not a probability assignment.

Because the exact boundary witness sets exclusive regions to zero, those empty regions carry no signed perturbation mass in the path-oscillation lemma. It establishes sharpness of the closed algebraic coordinate inequality, not by itself exact attainment by a common-support SxPID path. The valid full-support SxPID fixture below reaches more than 0.97 of the pointwise synergy misinformative bound and more than 0.95 of the pointwise synergy net bound. These are explicit near-attainment results, not an exact or global path-sharpness theorem.

It remains to control the weight terms. For net unique information, let a, b be the target-region masses and let $a_{\mathrm{f}} \geq a$, $b_{\mathrm{f}} \geq b$ be the full-event masses. Then

$$\exp(\pi^{\mathrm{sx}}_{\mathrm{unique}}) = \frac{a(a_{\mathrm{f}} + b_{\mathrm{f}})}{a_{\mathrm{f}}(a+b)} \geq \frac{4a}{(1+a)^2} \geq \frac{4p_{\min}}{(1+p_{\min})^2}. \tag{111}$$

The intermediate steps are

$$\frac{a(a_{\mathrm{f}} + b_{\mathrm{f}})}{a_{\mathrm{f}}(a+b)} \geq \frac{a(a_{\mathrm{f}} + b)}{a_{\mathrm{f}}(a+b)} \geq \frac{a}{(1-b)(a+b)}. \tag{112}$$

The first inequality uses $b_{\mathrm{f}} \geq b$. For the second, $a_{\mathrm{f}} + b_{\mathrm{f}} \leq 1$ and $b_{\mathrm{f}} \geq b$ imply $a_{\mathrm{f}} \leq 1 - b$; the function $(a_{\mathrm{f}} + b)/a_{\mathrm{f}} = 1 + b/a_{\mathrm{f}}$ decreases in a_{f}. Finally,

$$(1-b)(a+b) \leq \frac{((1-b) + (a+b))^2}{4} = \frac{(1+a)^2}{4}$$

by the arithmetic–geometric mean inequality. The function $4a/(1+a)^2$ is nondecreasing on $[0, 1]$. Under the p-law, this gives the $-J$ lower endpoint for each unique net atom. The upper endpoint is L because $N(a_{\mathrm{f}}, b_{\mathrm{f}}) \leq L$ and $N(a, b) \geq 0$. Moreover,

$$0 \leq \Phi(a, b, c) \leq J \qquad \text{under the } p\text{-law}. \tag{113}$$

For the upper bound, put $u = a + b + c \geq a$. The arithmetic-geometric mean inequality gives $(a+b)(a+c) \leq (a+u)^2/4$. The ratio $(a+u)^2/(4au)$ is nondecreasing in $u \geq a$, so $u \leq 1$ reduces it to $(1+a)^2/(4a)$. This last expression is nonincreasing on $(0,1]$, and $a \geq p_{\min}$ gives the upper endpoint J. Under the p-law, redundancy and unique net atoms lie in $[-J, L]$, and net synergy lies in $[-J, J]$. Every q-law cell has mass at least $p_{\min} - \eta$, so the same proof gives $0 \leq \Phi \leq J_q$ under q. These ranges control the p-weighted term in the average-change split and give the endpoint caps J_q and $J + J_q$. The other outer caps use support cardinality and therefore bound the averaged values under both laws. Centered total variation now gives the displayed atom-specific weight terms. This two-source result uses only the displayed log expressions and elementary range arguments. It does not use the published general-source component-nonnegativity result used in Theorem 6.3. □

The lower endpoint in Equation (67) is attained as an event bound when $p_{\min} \leq 1/3$, the intersection has mass $p_{\min}$, the two exclusive regions each have mass $(1-p_{\min})/2$, and there is no outside mass. A separate nested event of mass $p_{\min}$ and an outside cell of mass $1 - p_{\min}$ attains the upper endpoint when $p_{\min} \leq 1/2$. These separate endpoint laws do not prove that a general Möbius-row bound is sharp.

The revision-1 constants $(\Lambda, 2\Lambda, 3\Lambda)$ for cumulative terms and $(r_\alpha\Lambda, 2r_\alpha\Lambda, 3r_\alpha\Lambda)$ for pointwise atoms remain valid triangle-inequality baselines. They are superseded because they ignore the direct nested and intersection-PMI gradients. The retained generic two-point extremizer $f = (-r_\alpha L, r_\alpha L)$ is not shown to be SxPID-realizable and proves no SxPID sharpness claim.

For a realizable two-source family, in lexicographic binary-cell order let

$$p_\varepsilon = \left(\varepsilon, \frac{1}{6}, \frac{1}{3}, \varepsilon, \frac{1}{2} - 5\varepsilon, \varepsilon, \varepsilon, \varepsilon\right), \qquad 0 < \varepsilon < \frac{1}{12}. \tag{114}$$

Net synergy at 000 and 111 equals $-L + \log 10 + o(1)$ and $L - \log(40/3) + o(1)$. Their oscillation divided by L tends to two. Moving mass $0 < \rho < \varepsilon$ from 000 to 111 proves asymptotic necessity of the δL centered net-weight coefficient for that decomposition. For admissible $\rho < \varepsilon$, the Λ_{syn} pointwise-value term can dominate the total bound. Thus this family does not prove global sharpness of the complete average bound. Whether every nonbottom SxPID atom admits a stronger exact lower floor remains open.

The asymptotic statement is a prose derivation from the displayed family. The executable fixture checks only the finite case $\varepsilon = 1/60$. Its largest synergy misinformative and net changes are approximately $0.9806\Lambda_{\text{syn}}$ and $0.9597\Lambda_{\text{syn}}$. These ratios are bounded near-tightness evidence, not a global-sharpness proof.

Under the standing conditions $0 \leq \delta$, $0 < p_{\min} \leq 1$, and $\delta \leq p_{\min}/2$, put $\xi = \delta/(2p_{\min}) \leq 1/4$. The Lean theorem `refined_log_modulus_linearized_chain` proves

$$0 \leq \Lambda_{\text{syn}} \leq \Lambda \leq \frac{\xi}{1-\xi} \leq \frac{4\xi}{3} = \frac{2\delta}{3p_{\min}}. \tag{115}$$

Binary64 terminology and hypotheses. Let $\mathbb{F}_{64}$ denote the finite IEEE 754 binary64 values. The following branch argument assumes binary64 inputs, radix two, precision 53, round-to-nearest with ties to even, one binary64 rounding per basic arithmetic operation, no wider intermediate or excess-precision double rounding, and gradual underflow rather than flush-to-zero.

A positive *subnormal* has the form

$$k2^{-1074}, \qquad 1 \leq k < 2^{52}.$$

The least positive subnormal is 2^{-1074} and the least positive normal is 2^{-1022}. Gradual underflow means that correctly rounded operations retain the subnormal grid rather than replacing every result below 2^{-1022} by zero.

For a binary64 value with a finite predecessor, define

$$\mathrm{nextDown}(x) := \max\{y \in \mathbb{F}_{64} : y < x\}.$$

In particular,

$$\mathrm{nextDown}(1/2) = 1/2 - 2^{-54}.$$

Sterbenz's lemma states that if positive binary floating-point values x, y satisfy $x/2 \le y \le 2x$, then subtraction returns the exact real difference $x - y$, possibly as a subnormal when gradual underflow is enabled.

For $x > -1$, use the explicit convention

$$\mathrm{log1p}(x) = \log(1 + x), \qquad \mathrm{log1pmx}(x) = \mathrm{log1p}(x) - x.$$

Here a compensated `log1pmx` routine means one designed to evaluate the combined expression without naively subtracting two nearly equal rounded values. Consequently,

$$-\log(1 - r) - r = -\,\mathrm{log1pmx}(-r).$$

These assumptions justify only the stated binary64 branch reasoning. They do not assert correct rounding of transcendental library functions or provide a global floating-point error theorem.

For numerical evaluation, require finite inputs with $0 < p_{\min} \le 1$ and $0 \le \eta < p_{\min}$. First compute $q_{\mathrm{floor}} = p_{\min} - \eta$ in the working precision. Reject the input if this value is not finite and strictly positive. One formula is not stable over the complete strict-support range. When $p_{\min}$ and q_{floor} are subnormal and $q_{\mathrm{floor}} \ll p_{\min}$, the rounded ratio $\eta/p_{\min}$ can lose most of the remaining-floor information.

If $r = \eta/p_{\min} \le 1/2$, evaluate $\Lambda = -\log(1 - r)$ as `-log1p(-r)` or as $r + \sum_{k=2}^{\infty} r^k/k$. The committed test-only evaluator uses the positive-series form. Do not subtract nearby binary64 values. Use

$$\Lambda_{\mathrm{syn}} = [-\log(1 - r) - r] + r(1 - p_{\min}). \tag{116}$$

Evaluate the bracket as $-\,\mathrm{log1pmx}(-r)$ with a compensated routine, or by the positive series $\sum_{k=2}^{\infty} r^k/k$. The second term avoids both reciprocal overflow and the cancellation in $\eta(1/p_{\min} - 1)$.

If $r > 1/2$, use

$$u = \frac{q_{\mathrm{floor}}}{p_{\min}}, \qquad \Lambda = -\log u, \qquad \Lambda_{\mathrm{syn}} = \Lambda - \eta. \tag{117}$$

Use $q_{\mathrm{floor}}/p_{\min}$ directly. Under this bounded input contract, $p_{\min}/q_{\mathrm{floor}} \le 2^{53}$, so the inverse quotient cannot overflow. The direct quotient keeps the documented normal interval and avoids an unnecessary inversion. Do not subtract two large logarithms when $p_{\min}$ is very small, because that route can lose relative accuracy. For binary64 inputs on this upper branch, $p_{\min}/2 < \eta < p_{\min}$. Sterbenz subtraction therefore makes $q_{\mathrm{floor}} = p_{\min} - \eta$ exact. The smallest positive represented gap gives $u \ge 2^{-53}$, so u is normal. IEEE binary64 round-to-nearest, ties-to-even division keeps the computed quotient below $1/2$ for the contracted inputs. The bounded evaluator enforces $2^{-53} \le u \le \mathrm{nextDown}(1/2)$ and does not accept $u = 1/2$. To see the strict result, upper-branch selection implies the exact ratio $\eta/p_{\min} > 1/2 + 2^{-54}$; a midpoint tie would round to $1/2$. Exact subtraction then gives $q_{\mathrm{floor}}/p_{\min} < 1/2 - 2^{-54}$ before the second division is rounded. This argument uses the binary64 conventions above. The committed bounded test checks the quotient precondition and uses runtime canaries for the rounding mode and gradual underflow before it uses the quotient-log route. The generator converts the upper-branch inputs to exact binary fractions. It rejects a represented floor that is not exactly $p_{\min} - \eta$. This binary64 argument is numerical guidance, not part of the exact-real theorem. The branch at $r = 1/2$ is an implementation choice.

Evaluate the endpoint ceiling without first forming its quotient. If $q_{\text{floor}} \le 1/2$, use

$$J_q = 2\log(1+q_{\text{floor}}) - \log q_{\text{floor}} - \log 4. \tag{118}$$

If $q_{\text{floor}} > 1/2$, put $t = (1-q_{\text{floor}})/(1+q_{\text{floor}})$ and use

$$J_q = -\log(1-t^2) \tag{119}$$

with a `log1p` operation. The first route avoids overflow for a subnormal floor. The second avoids cancellation as the floor tends to one.

The committed generator uses 400-digit Decimal arithmetic. It evaluates Λ with one high-precision quotient logarithm. The Rust reconstruction is a bounded fixture check for the committed laws and numerical stress cases, not a general interval implementation. No result in this section is a binary64 error theorem, a support-creation theorem, or a continuous-estimator theorem.

7 Composition with SxPID

For common-law sampling, empirical support is a subset of population support almost surely. On the event in Theorem 4.1, the strict condition

$$D_n < 2p_{\min} \tag{120}$$

forces empirical and population support to agree. Indeed, if a population cell z were absent empirically, its negative deviation would be at least $p_{\min}$; because the signed deviations sum to zero, their positive part has the same mass, and the L^1 distance would be at least $2p_{\min}$. The realized error satisfies $\delta \le D_n$. Both pointwise moduli are nondecreasing in δ. Explicitly,

$$\frac{\partial \Lambda}{\partial \delta} = \frac{1}{2(p_{\min} - \delta/2)} > 0, \tag{121}$$

and

$$\frac{\partial \Lambda_{\text{syn}}}{\partial \delta} = \frac{1}{2(p_{\min} - \delta/2)} - \frac{1}{2} \ge 0. \tag{122}$$

The last inequality holds because the remaining floor $p_{\min} - \delta/2$ is positive and at most one. For $f \in (0,1]$, the endpoint function $J(f) = 2\log(1+f) - \log f - \log 4$ has derivative $(f-1)/(f(1+f)) \le 0$. Since the floor $f = p_{\min} - \delta/2$ decreases with δ, J_q is nondecreasing. Sums and minima of these nondecreasing nonnegative expressions remain nondecreasing. Thus every displayed right-hand side is nondecreasing. Substitute D_n for δ everywhere, including in η, Λ_{syn}, and J_q. Use the result in Equations (52), (59), (60), (61), and (71), (72), together with the two-source table. This gives simultaneous exact-real envelopes for every supported cumulative term, every fixed general-source Möbius atom, and every complete two-source atom.

A usable numerical envelope needs a justified lower bound on $p_{\min}$. The observed minimum frequency is not such a bound. A positive population cell can be absent from a finite sample.

For Theorem 5.1, also require every P_i to be supported inside $\operatorname{supp}(P_\star)$. Define

$$p_{\min,\star} = \min_{z \in \operatorname{supp}(P_\star)} P_\star(z). \tag{123}$$

If

$$b_n + D_n < 2p_{\min,\star}, \tag{124}$$

then the realized error is at most $b_n + D_n$. Monotonicity gives the same composition with $b_n + D_n$ on the right-hand side.

Proposition 7.1 (Frozen held-out finite transform)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{G} \subseteq \mathcal{F}$ be the sigma-field generated by the training sample, all training randomness, and the fitted artifact. Let $F_{\mathrm{fit}} \in \mathcal{G}$ be the event that fitting does not return a usable total transform, and impose the zero-failure condition*

$$\mathbb{P}(F_{\mathrm{fit}}) = 0. \tag{125}$$

Let $(\mathsf{X}, \mathcal{A})$ be a standard-Borel raw-row space, let μ be a probability law on it, and let $X_1, X_2, \ldots$ be i.i.d. with law μ, with $\sigma(X_i : i \geq 1)$ independent of $\mathcal{G}$. Let $\mathcal{Z}$ be one fixed nonempty finite output alphabet; when invoking SxPID, its elements are the complete source–target symbols introduced in Section 2. On $F_{\mathrm{fit}}^{\mathsf{c}}$, the fitted artifact determines a total finite-output map. Extend it arbitrarily by one fixed output on F_{fit}, and require the resulting joint map

$$\Phi : (\Omega \times \mathsf{X}, \mathcal{G} \otimes \mathcal{A}) \longrightarrow (\mathcal{Z}, 2^{\mathcal{Z}}) \tag{126}$$

to be measurable. Put $\Phi_\omega(x) = \Phi(\omega, x)$, $Z_i(\omega) = \Phi_\omega(X_i(\omega))$, and

$$P_\omega(z) = \mu\Big(\Phi_\omega^{-1}(\{z\})\Big), \qquad z \in \mathcal{Z}. \tag{127}$$

Then $P_\omega = \Phi_{\omega\#}\mu$ is a $\mathcal{G}$-measurable probability kernel and, for every $k \geq 1$, all distinct $i_1, \ldots, i_k$, and all $z_1, \ldots, z_k \in \mathcal{Z}$,

$$\mathbb{P}(Z_{i_1} = z_1, \ldots, Z_{i_k} = z_k \,|\, \mathcal{G}) = \prod_{\ell=1}^{k} P_\omega(z_\ell) \quad a.s. \tag{128}$$

Thus the transformed evaluation rows are conditionally i.i.d. with the conditional pushforward law P_ω. With $\widehat{P}_n$ defined from these Z_i as in Section 2, all of the following hold on one event of conditional probability one:

$$\mathbb{P}\left(\begin{array}{c} \forall z \in \mathcal{Z} : \ \widehat{P}_n(z) \longrightarrow P_\omega(z), \\ \forall n \geq 1 : \ \mathrm{supp}(\widehat{P}_n) \subseteq \mathrm{supp}(P_\omega), \\ \exists N < \infty \ \forall n \geq N : \ \mathrm{supp}(\widehat{P}_n) = \mathrm{supp}(P_\omega) \end{array} \middle| \ \mathcal{G} \right) = 1 \quad a.s. \tag{129}$$

Proof. Joint measurability in Equation (126) and the parameter-integral theorem make each cell mass in Equation (127) $\mathcal{G}$-measurable. The masses are nonnegative and sum to one. Independence of the complete evaluation sequence from $\mathcal{G}$ means that the conditional law of every finite vector $(X_{i_1}, \ldots, X_{i_k})$ is μ^k. Integrating the product of the finite-alphabet cell indicators

$$\prod_{\ell=1}^{k} \mathbf{1}\{\Phi_\omega(X_{i_\ell}) = z_\ell\}$$

against μ^k proves Equation (128).

For each fixed $z \in \mathcal{Z}$, conditional Hoeffding gives, for every $\varepsilon > 0$,

$$\mathbb{P}\Big(|\widehat{P}_n(z) - P_\omega(z)| \geq \varepsilon \,\Big|\, \mathcal{G}\Big) \leq 2e^{-2n\varepsilon^2} \quad \text{a.s.} \tag{130}$$

The right-hand side is summable in n. Conditional Borel–Cantelli, first over positive rational ε and then over the finite alphabet, gives the first line of Equation (129) on one common conditional-probability-one event.

For a cell with $P_\omega(z) = 0$, the conditional union bound gives

$$\mathbb{P}(\exists i \geq 1 : Z_i = z \mid \mathcal{G}) = 0 \quad \text{on } \{P_\omega(z) = 0\}. \tag{131}$$

This proves simultaneous support containment after another finite intersection. On $\{P_\omega(z) > 0\}$, conditional independence gives

$$\mathbb{P}(Z_1 \neq z, \ldots, Z_N \neq z \mid \mathcal{G}) = (1 - P_\omega(z))^N \longrightarrow 0. \tag{132}$$

Continuity from above therefore shows that every positive conditional-law cell occurs at least once. Because $\mathcal{Z}$ is finite, the maximum of those first-occurrence indices is finite, and support equality persists thereafter. Equation (125) makes the arbitrary extension on the fitting-failure event irrelevant to every almost-sure conclusion. □

For $|\mathcal{Z}| \geq 2$, conditional on $\mathcal{G}$, Theorems 3.1 and 4.1 apply with one occupied color and $V_n = n$. The finite random quantity

$$p_{\min}(\omega) = \min_{z \in \operatorname{supp}(P_\omega)} P_\omega(z) > 0 \tag{133}$$

is a conditional support floor. Equation (129) implies $\left\|\widehat{P}_n - P_\omega\right\|_1 \to 0$ and hence eventual satisfaction of the strict margin $\left\|\widehat{P}_n - P_\omega\right\|_1 < 2p_{\min}(\omega)$. The local moduli above then give simultaneous exact-real plug-in consistency for the displayed finite family of supported cumulative terms and SxPID atoms, with conditional target P_ω. A training-measurable random floor gives a training-conditional numerical envelope. A deterministic numerical envelope uniform over training artifacts requires a justified deterministic lower bound for $p_{\min}(\omega)$.

The unconditional one-row law is the mixture

$$P_{\text{mix}}(z) = \mathbb{P}(Z_i = z) = \mathbb{E}[P_\omega(z)]. \tag{134}$$

The shared fitted artifact generally makes the Z_i dependent without conditioning, and the empirical law in Proposition 7.1 converges to the random component P_ω, not to P_{mix}. Consequently the proposition does not establish consistency for SxPID evaluated at the unconditional mixture; that interpretation is valid only under an additional condition such as $P_\omega = P_{\text{mix}}$ almost surely. In general, nonlinearity also prevents interchanging the SxPID functional with the mixture expectation.

The proposition is the held-out i.i.d. case and uses one color. For a conditionally dependency-colored extension, the color map must be fixed after conditioning on $\mathcal{G}$, and the common conditional row law and mutual independence of all complete rows inside every color must be proved. Theorems 3.1 and 4.1, and their stated sufficient condition for convergence, can then be applied conditionally. A positive-probability fitting failure, a transform selected from evaluation outcomes, or merely pairwise conditional independence does not meet this contract.

The displayed common-law envelope proves almost-sure exact-real finite-alphabet SxPID plug-in consistency under the sufficient condition $V_n \log(n)/n^2 \to 0$; fixed d is sufficient. Indeed, that condition makes $D_n \to 0$. For every $\alpha \in (0, 1)$, Theorem 4.1 then places convergence on an event of probability at least $1 - \alpha$; letting $\alpha \downarrow 0$ through a countable sequence makes the convergence event probability one. Equivalently, the fixed-threshold tails are summable and Borel–Cantelli applies. The displayed drift envelope proves almost-sure exact-real reference-law SxPID consistency when this sufficient condition holds and $b_n \to 0$. These are not necessary conditions under a stronger sampling theorem. The result does not validate continuous $I_\cap^{\text{sx}}$, arbitrary overlapping windows, adaptive preprocessing on evaluation rows, unknown mixing rates, or binary64 asymptotics.

8 Counterexamples and invalidated routes

Invalid route	Counterexample	Required correction
Pairwise independence inside one color	Let U be uniform on $\mathbb{F}_2^4$. Use the 15 nonzero linear forms $a \cdot U$ as binary rows. Every pair is independent. All rows are zero with probability $1/16$. The empirical L^1 error is then one, while the false one-color bound is $2e^{-15/2} \approx 0.00111$.	Require mutual independence of all complete rows in each color.
Remove the color factor	Take m independent fair bits and copy each bit once into each of d colors. Each color is i.i.d., but only $m = n/d$ independent values remain.	Keep V_n, or keep the coarser factor d.
Infer convergence from one color per row	Set every row equal to one common fair bit. Singleton colors satisfy their local premise, but $V_n = n^2$ and the empirical law does not converge to the fair law.	This coloring cannot prove convergence; use a stronger coloring or another theorem.
Read the envelope condition as necessary	Let the rows be i.i.d. but assign each row a separate color. Then $V_n = n^2$, although the empirical law converges by the strong law.	State $V_n \log(n)/n^2 \to 0$ only as a sufficient condition for this envelope.
Data-adaptive coloring without the complete conditional contract	Set every row equal to one fair bit, then select one occupied color from that bit. Conditional rows are degenerate, but their conditional law is not the common unconditional law.	Require fixed colors, or prove the full conditional law and independence premises.
Substitute a mixing label for the color premise	Use a stationary fair two-state Markov chain with flip probability $\theta = 1/100$. Its transition dependence decays geometrically. For two rows, the empirical L^1 error is one with probability $1 - \theta = 0.99$, above the false one-color bound $2e^{-1}$.	Use a theorem with an explicit mixing coefficient and rate, or prove a valid independence coloring.
Pairwise lag independence as strong ℓ-dependence	The finite-field construction has pairwise independence and a joint failure.	Require the sigma-field definition. Residues modulo $\ell + 1$ then give valid colors by ordered induction.
Halve the net weight term from the generic range alone	On two points, let $w = (-\delta/2, \delta/2)$ and a generic row value be $f = (-r_\alpha L, r_\alpha L)$. Then $\lvert \sum wf \rvert = r_\alpha L\delta$. This row value is not shown to be realizable by an SxPID atom when $r_\alpha > 1$. It proves no SxPID sharpness claim.	Keep the generic range baseline unless a separate SxPID argument supplies a smaller range.

Invalid route	Counterexample	Required correction
Replace Λ by Λ_{syn} for every two-source atom	The displayed two-cell SxPID laws make a redundancy or unique component change by exactly $\Lambda > \Lambda_{\mathrm{syn}}$ when $\eta > 0$.	Use Λ_{syn} only for synergy. Keep Λ for redundancy and unique information.
Allow $\delta = 2p_{\min}$	Move all mass from a least-probable supported cell. Its required logs become undefined.	Keep $\delta < 2p_{\min}$ strict.
Control only separate marginals	Set $S_1 = S_2 = X$. For $0 < \rho < 1$, at the supported realization $(X, T) = (0, 0)$, use fair binary X, T marginals with diagonal cells $(1 + \rho)/4$ and off-diagonal cells $(1 - \rho)/4$, then replace ρ by $-\rho$. Marginals stay fixed, but pointwise redundancy changes by $\log((1 + \rho)/(1 - \rho))$.	Control the complete joint law.
Allow new support with a linear average bound	Let $p(0, 0, 0) = 1$, and let $q(0, 0, 0) = 1 - \varepsilon$, $q(1, 1, 1) = \varepsilon$. Then $\lVert q - p \rVert_1 = 2\varepsilon$, while the averaged redundancy is $-(1 - \varepsilon) \log(1 - \varepsilon) - \varepsilon \log \varepsilon$.	Require common support or use a nonlinear boundary modulus.
Use observed minimum frequency as a population floor	An unobserved positive cell has empirical frequency zero.	Use an external floor or a separate simultaneous lower-confidence result.

The executable challenge suite enumerates the finite-field pairwise-independence construction, copied-color construction, singleton-color construction, and adaptive-color construction. It also checks a support-deletion boundary, an unspecified-mixing construction, a generic net-weight range extremizer, a univariate-marginal-only construction, and a new-support construction. Three cases show that endpoint validity does not replace the componentwise nonnegative lift premise. Seven rational cases check the eight conditioned-diamond coordinates and all 64 ordered differences per case. They include a zero-lift boundary and one explicitly algebra-only, unnormalized case. Nine cases have six positive displayed masses that sum to one and realize all nine exact minimum/maximum regimes. The same exact inputs also check the ordinary-diamond identity and the conditioned-nested candidate and closed-form diameter identities. Six two-cell SxPID cases attain Λ for non-synergy components and violate the false all-atom refinement. The remaining checks cover the telescoping allocation, class-size inequalities, high-precision one-log cases, and all displayed bounds on six committed two-source law pairs. Two pairs make the J_q component cap and the $J + J_q$ net cap active. Ten binary64 cases challenge the adaptive refined-modulus evaluation. Stored hexadecimal payloads bind parsed operands and subtraction results. The cases include adjacent inputs around $r = 1/2$, the exact seam, the exact lower endpoint $q_{\mathrm{floor}}/p_{\min} = 2^{-53}$ of the upper route, an extreme normal-scale input, and a subnormal near-boundary floor. Six cases challenge the endpoint-ceiling evaluation. They include adjacent inputs around $q_{\mathrm{floor}} = 1/2$, the exact seam, a floor near one, the exact $q_{\mathrm{floor}} = 1$ positive-zero endpoint, and the smallest positive subnormal floor.

Their 400-digit references use the exact real values represented by the parsed binary64 inputs. The generator stores the $q_{\mathrm{floor}} = 1$ endpoint as the exact algebraic value zero. It does not derive that endpoint by subtracting rounded logarithms. The bounded evaluator also returns this endpoint directly. It does not depend on the signed-zero result of a logarithm call. One full-support pair has count vectors

$$(10, 100, 200, 10, 250, 10, 10, 10) \quad \text{and} \quad (1, 100, 200, 10, 250, 10, 10, 19). \tag{135}$$

It has $\delta = 3/100$, $\eta = 3/200$, $p_{\min} = 1/60$, and $p_{\min}/(p_{\min} - \eta) = 10$. Its largest observed synergy misinformative and net changes exceed $0.98\Lambda_{\mathrm{syn}}$ and $0.959\Lambda_{\mathrm{syn}}$. This is bounded near-tightness evidence for the synergy pointwise constant, not a proof of global sharpness.

The fixed-window fixture uses independent fair innovations U_t, V_t and

$$S_{1,t} = U_t \text{ OR } U_{t+1}, \qquad S_{2,t} = V_t \text{ OR } V_{t+1}, \qquad T_t = S_{1,t} \text{ XOR } S_{2,t}. \tag{136}$$

Even and odd starts are the two colors. Its exact joint count weights are $1 : 3 : 3 : 9$. The high-precision Decimal oracle supplies expected logarithmic values. For stored logarithmic constants and bounds, the Rust reconstruction tolerance is $32\varepsilon_{\mathrm{mach}} \max(1, |x_{\mathrm{Rust}}|, |x_{\mathrm{oracle}}|)$. For categorical estimator outputs, the absolute comparison ceiling is $32\varepsilon_{\mathrm{mach}}$ nats, where $\varepsilon_{\mathrm{mach}} = 2^{-52}$ for binary64. The test also checks that row reversal gives bit-identical output.

9 Formal and executable evidence boundary

The Lean project is pinned to one toolchain and one exact dependency manifest. The repository checker rejects admitted proof placeholders, builds the project, and replays its declarations with Lean's kernel checker. It enforces an exact ordered inventory of all 263 source-level declarations across the seven imported modules and audits the permitted axiom basis of all 201 named source theorems. The complete two-source count/event bridge is SHA-256 bound. A separately digest-pinned and compiled semantic contract fixes 16 paper-facing event, count, nonnegativity, positivity, fractional-cover, and generic Möbius claims; those examples are not individually passed through `collectAxioms`. Its self-test rejects ten static mutations, including valid same-name theorem weakening, and five baseline-first isolated Lean semantic mutations under normal and optimized Python.

The checked Lean project proves the following results across the `Dependence.lean` and `LocalContinuity.lean` source files:

- the event-mass $\delta/2$ lemma and exact attainment by the positive-coordinate event;
- positivity under the strict support margin;
- the one-log modulus, its monotonicity, and its rational simplification;
- the centered zero-sum oscillation inequality;
- reciprocal-diameter bounds for negative-event-log, nested-log-ratio, and intersection-PMI gradient coordinates;
- the exact ordinary two-source diamond gradient diameter, its attainment, its union-reciprocal corollary, and the algebraic and logarithmic $0 \le \Phi \le J$ bounds;
- the five-coordinate conditioned nested-event candidate extrema, exact closed-form gradient diameter, attainment, zero-side-mass algebraic witness, and impossibility of a positive uniform subtraction on that algebraic domain;
- the exact candidate-extrema form and the union-reciprocal upper bound for the eight-coordinate conditioned-diamond gradient, plus its normalized probability-domain corollaries, under one positive common mass and nonnegative exclusive and complement-region masses; an exact-real boundary witness establishes sharpness of the closed algebraic coordinate bound, while valid full-support SxPID fixtures establish bounded near-attainment rather than exact path sharpness;

- the complete small-error linearization chain for Λ_{syn} under $0 \le \delta \le p_{\min}/2$ and $0 < p_{\min} \le 1$;
- the effective-color proxy bounds and the normalized factor range;
- the generic absolute linear-row bound;
- component and net range transfer, centered weight bounds, and combined finite-average perturbation bounds;
- the factorized positive segment-event floor;
- the exact unit-scale finite telescoping allocation; and
- the algebraic radius-exponent cancellation.

Lean formalizes the finite heterogeneous keyed Sx event layer and generic fractional-cover event corollaries. It also binds supplied exact natural counts with positive total to the four fixed two-source signed-net cumulative logarithms and their positive-support averages. It does not connect those event masses to the derivative-shaped coordinates in `LocalContinuity`, formalize differentiation or path integration, define and invert concrete Möbius atoms, import the published component-sign result, or prove the complete informative or misinformative averaged component family. Lean also does not encode random variables or conditional laws, generalized Hölder, Hoeffding's lemma, the Chernoff argument, the probability union bound, the drift result, or the fixed-window independence argument. Bytes or rows to counts, Rust or floating-point refinement, certifier/parser execution, and sampling, population, calibration, and consumer validity remain outside Lean. The probability theorem is a prose proof; Lean machine-checks only the deterministic obligations listed above. The Python suite uses exact rational arithmetic for finite probability and count identities, plus 400-digit Decimal arithmetic for logarithms and an absolute 10^{-350} tolerance for its internal Decimal identity checks. It checks all 64 ordered conditioned-diamond coordinate pairs in each of seven rational cases, all nine exact extremal regimes, and three endpoint-valid negative-lift counterexamples. Transcendental logarithms are not exact rational values. From each local count-table pair, the Rust test independently reconstructs both empirical laws, δ, η, $p_{\min}$, Λ, Λ_{syn}, L, h, J, J_q, the atom family, and every stored bound before it checks the implementation outputs. A separate bounded test reconstructs the ten adaptive-modulus and six endpoint-ceiling numerical cases and requires selected naive routes to fail. For this stability test only, an expected zero must match positive zero bit for bit. Each other comparison uses

$$256\varepsilon_{\text{mach}} \max(|x_{\text{oracle}}|, s_{\text{op}}), \tag{137}$$

where s_{op} is the bounded sum of the absolute displayed operation terms for that case. A selected naive route must be nonfinite or differ from the oracle by more than 1024 times that tolerance. The separate law reconstruction and categorical-output tolerances remain the $32\varepsilon_{\text{mach}}$ rules declared above. These checks are bounded executable evidence. They are not a general proof, a global binary64 bound, or external review.

10 Ecosystem use boundary

Consumer	Result supplied here	Still required
Prisoma	Categorical SxPID law control after a frozen finite map when held-out rows meet the color contract	Concrete row law, justified support floor, independent transform receipt, and a valid window coloring
Galadriel	Dependence-aware categorical alternative to an i.i.d. plug-in claim	Removal of same-window fitting and generic jitter routes; explicit observation and coloring contracts
Haldir	Mathematical target for block or residue-class designs	Direct dependency, corrected atom interpretation, and proof of strong ℓ-dependence or another valid coloring
Crebain	Finite-alphabet validation component for a future integration	Exact observation mapping, run-log contract, direct dependency, and explicit scientific estimand

This document does not assert that a consumer meets these premises. Integration remains `not_claimed` until the consumer records and checks them.

11 Reproduction

From the repository root, run

```
python3 scripts/check-lean-finite-convergence.py
python3 scripts/generate-dependency-colored-sxpid-oracle.py
cargo test --locked -p pid-core --test dependency_colored_sxpid_oracle
scripts/check-dependency-colored-sxpid-pdf.sh
```

The PDF checker rebuilds this source with a fixed source-date epoch, rejects LaTeX diagnostics, and compares the exact PDF bytes. The Lean checker binds the toolchain, manifest, source set, dependency revisions, dependency origins, and clean checkout state. It removes ambient Git routing and global or system configuration while retaining each dependency checkout's local origin configuration for verification.

12 Conclusion

The result supplies a dependence-aware finite-sample path from a declared finite-row coloring to exact-real categorical shared-exclusions PID bounds. Its displayed envelope proves almost-sure exact-real consistency under the sufficient condition $V_n \log(n)/n^2 \to 0$. The corresponding reference-law result also uses $b_n \to 0$. These are not necessary conditions under a stronger sampling theorem. The result separates four objects that must remain separate: sampling fluctuation, deterministic drift, support margin, and program arithmetic. The color-size proxy improves the proof constant for unbalanced classes. The telescoping allocation gives a simple time-uniform statement. The local logarithmic modulus gives a direct law-to-atom transfer once common support is justified. Every weaker route listed in Section 8 is retained as invalidated work.

References

[1] A. Makkeh, A. J. Gutknecht, and M. Wibral. Introducing a differentiable measure of pointwise shared information. *Physical Review E*, 103:032149, 2021. https://doi.org/10.1103/PhysRevE.103.032149.

[2] A. J. Gutknecht, M. Wibral, and A. Makkeh. Bits and pieces: understanding information decomposition from part-whole relationships and formal logic. *Proceedings of the Royal Society A*, 477:20210110, 2021. https://doi.org/10.1098/rspa.2021.0110.

[3] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, 1963. https://doi.org/10.1080/01621459.1963.10500830.

[4] S. Janson. Large deviations for sums of partly dependent random variables. *Random Structures & Algorithms*, 24(3):234–248, 2004. https://doi.org/10.1002/rsa.20008.

[5] C. Pelekis, J. Ramon, and Y. Wang. Hölder-type inequalities and their applications to concentration and correlation bounds. *Indagationes Mathematicae*, 28(1):170–182, 2017. https://doi.org/10.1016/j.indag.2016.11.017.

[6] T. Weissman, E. Ordentlich, G. Seroussi, S. Verdú, and M. J. Weinberger. Inequalities for the L^1 deviation of the empirical distribution. HPL-2003-97 (R.1), 2003. https://shiftleft.com/mirrors/www.hpl.hp.com/techreports/2003/HPL-2003-97R1.pdf.